

Physics 827: Homework Set No. 6

Due date: Friday, November 11, 2011, 1:00pm
in PRB M2043 (Biao Huang's office)

Total point value of set: 55 points

Problem 1 (10 pts.): Exercise 5.2.1 (Shankar, p. 163)

Problem 2 (20 pts.): Exercise 5.2.2 (Shankar, p. 163). *Hint:* I found it useful to replace the potential $V(x)$ by a suitable Gaussian potential of smaller magnitude in order to put an upper bound on the energy expectation value.

Problem 3 (5 pts.): Exercise 5.3.1 (Shankar, p. 166)

Problem 4 (5 pts.): Exercise 5.3.4 (Shankar, p. 167)

Problem 5 (15 pts.):

(a) (5 pts.) According to Postulate II, the angular momentum operator is defined as $\hat{\vec{L}} = \hat{\vec{X}} \times \hat{\vec{P}}$. Is this operator Hermitean? If not, write down a Hermitean expression for the angular momentum.

(b) (10 pts.) Using Ehrenfest's theorem, derive the equation of motion for the expectation value of angular momentum in an arbitrary state $|\psi\rangle$ for a particle in a 3-dimensional harmonic oscillator potential (with Hamiltonian given by Eq. (4.3.4) in Shankar).

Exercise 5.2.1 (page 163)

The probability amplitude ^{in position space} ψ for a particle in the ground state of a box of length L is given by the ground state wavefunction

$$\psi_1(x) = \begin{cases} \sqrt{\frac{2}{L}} \cos\left(\frac{\pi x}{L}\right) & (|x| \leq \frac{L}{2}) \\ 0 & (|x| > \frac{L}{2}) \end{cases} \quad \text{(if we put the box symmetrically around } x=0 \text{).}$$

The ground state for a box of length $2L$ is

$$\phi_1(x) = \begin{cases} \sqrt{\frac{1}{L}} \cos\left(\frac{\pi x}{2L}\right) & , |x| \leq L \\ 0 & |x| \geq L \end{cases}$$

The probability to find the particle described by $|\psi_1\rangle$ in the state $|\phi_1\rangle$ is

$$P = |\langle \phi_1 | \psi_1 \rangle|^2 = \left(\int_{-L/2}^{L/2} \sqrt{\frac{2}{L^2}} \cos\left(\frac{\pi x}{L}\right) \cos\left(\frac{\pi x}{2L}\right) dx \right)^2$$

(the inner product receives contributions only from $-\frac{L}{2} < x < \frac{L}{2}$ where $\psi_1(x) \neq 0$.)

To compute the overlap integral, substitute $\frac{\pi x}{L} = \xi$

($-\frac{\pi}{2} \leq \xi \leq \frac{\pi}{2}$, $dx = \frac{L}{\pi} d\xi$):

$$\sqrt{\frac{2}{L^2}} \int_{-L/2}^{L/2} dx \cos\left(\frac{\pi x}{L}\right) \cos\left(\frac{\pi x}{2L}\right) = \frac{\sqrt{2}}{\pi} \int_{-\pi/2}^{\pi/2} d\xi \cos(\xi) \cos(\xi/2)$$

$$= \frac{\sqrt{2}}{\pi} \int_{-\pi/2}^{\pi/2} d\xi \frac{1}{2} (\cos(\xi/2) + \cos(3\xi/2)) = \frac{1}{\pi\sqrt{2}} \left[2 \sin(\xi/2) + \frac{2}{3} \sin(3\xi/2) \right]_{-\pi/2}^{\pi/2}$$

$$= \frac{\sqrt{2}}{\pi} \left(2 \cdot \frac{1}{\sqrt{2}} + \frac{1}{3} \cdot 2 \cdot \frac{1}{\sqrt{2}} \right) = \frac{2\sqrt{2}}{\pi} \frac{4}{3\sqrt{2}} = \frac{8}{3\pi}$$

($\sin \frac{\pi}{4} = \sin(\frac{3\pi}{4}) = \frac{1}{\sqrt{2}}$)

$$\Rightarrow \boxed{P = \left(\frac{8}{3\pi}\right)^2} \checkmark$$

Exercise 5.2.2 (page 163)

- (a) Show that if $\langle \psi | \psi \rangle = 1$, $\langle \psi | \hat{H} | \psi \rangle \geq E_0$ where E_0 is the lowest energy eigenvalue.

Expand $|\psi\rangle$ in energy eigenstates $|E\rangle$, $\hat{H}|E\rangle = E|E\rangle$:

$$|\psi\rangle = \sum_E \langle E | \psi \rangle |E\rangle \equiv \sum_E a_E |E\rangle$$

$$\Rightarrow \langle \psi | \hat{H} | \psi \rangle = \sum_E \sum_{E'} a_E a_{E'}^* \underbrace{\langle E' | \hat{H} | E \rangle}_{E \delta(E, E')}$$

$$= \sum_E |a_E|^2 E \geq \sum_E |a_E|^2 E_0 \text{ since } E \geq E_0 \forall E.$$

$$\text{But since } \langle \psi | \psi \rangle = 1 = \sum_E \sum_{E'} a_E a_{E'}^* \underbrace{\langle E' | E \rangle}_{\delta(E, E')} = \sum_E |a_E|^2$$

this shows that

$$\langle \psi | \hat{H} | \psi \rangle \geq E_0 \sum_E |a_E|^2 = E_0 \quad (\text{Q.E.D.})$$

- (b) Thm: Every attractive potential in one dimension has at least one bound state.

Proof: Normalize $V(\infty) = 0$ such that $V(x) = -|V(x)| \leq 0$ for all x . Also, we can shift x such that $V(x)$ is minimal at $x=0$.

Consider the trial wave function

$$\psi_\alpha(x) = \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2/2} \quad \left(\int_{-\infty}^{\infty} |\psi_\alpha(x)|^2 dx = 1 \right)$$

which has one adjustable real parameter α .

Calculate

$$E(\alpha) = \langle \psi_\alpha | \hat{H} | \psi_\alpha \rangle = \langle \psi_\alpha | \frac{\hat{p}^2}{2m} - |V(x)| | \psi_\alpha \rangle$$

$$\stackrel{(x\text{-basis})}{=} \int_{-\infty}^{\infty} dx \psi_\alpha^*(x) \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} - |V(x)| \right) \psi_\alpha(x)$$

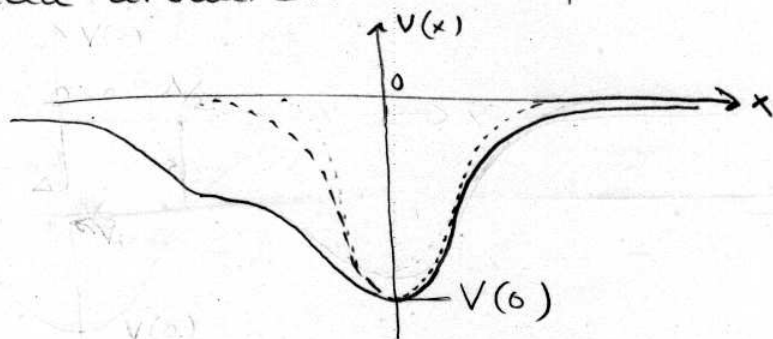
$$\frac{d\psi_\alpha}{dx} = \left(\frac{\alpha}{\pi} \right)^{1/4} (-\alpha x) e^{-\alpha x^2/2}$$

$$\frac{d^2\psi_\alpha}{dx^2} = \left(\frac{\alpha}{\pi} \right)^{1/4} \left(-\alpha e^{-\alpha x^2/2} + \alpha^2 x^2 e^{-\alpha x^2/2} \right)$$

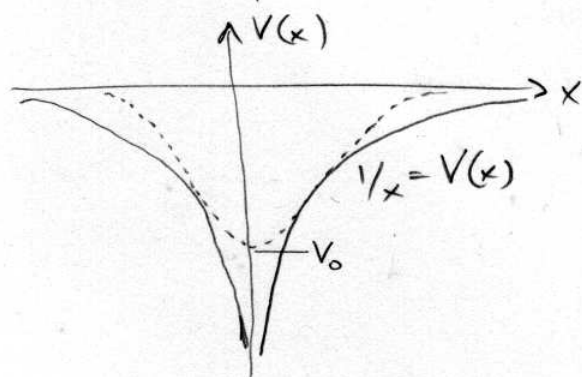
$$\begin{aligned} \Rightarrow E(\alpha) &= \sqrt{\frac{\alpha}{\pi}} \left[- \int_{-\infty}^{\infty} dx e^{-\alpha x^2} |V(x)| + \alpha \frac{\hbar^2}{2m} \underbrace{\int_{-\infty}^{\infty} dx e^{-\alpha x^2}}_{\sqrt{\frac{\pi}{\alpha}}} - \alpha^2 \frac{\hbar^2}{2m} \underbrace{\int_{-\infty}^{\infty} dx x^2 e^{-\alpha x^2}}_{\frac{1}{2\alpha} \sqrt{\frac{\pi}{\alpha}}} \right] \\ &= -\sqrt{\frac{\alpha}{\pi}} \int_{-\infty}^{\infty} dx e^{-\alpha x^2} |V(x)| + \frac{\alpha \hbar^2}{4m} \end{aligned}$$

So the energy $\langle V \rangle_\alpha + \langle T \rangle_\alpha$ is the sum of a negative potential energy term and positive kinetic energy contribution. The kinetic energy decreases linearly with α as $\alpha \rightarrow 0$ (i.e. as the wave function spreads out).

The potential energy $\langle V \rangle_\alpha$ decreases monotonically with increasing $|V(x)|$. So we can put an upper bound on the potential energy by replacing the potential by a Gaussian around $x=0$ that "fits inside the potential":



If $V(0)$ is infinite, we can still fit a finite Gaussian inside the potential such as follows:



in this case $V_0 \neq V(0)$, $|V_0| < |V(0)|$

We thus have

$$E(\alpha) \leq -\sqrt{\frac{\alpha}{\pi}} |V_0| \int_{-\infty}^{\infty} dx e^{-\alpha x^2} e^{-x^2/\Delta^2} + \frac{\alpha \hbar^2}{4m}$$

Where Δ is the width of the Gaussian potential and $|V_0|$ is its depth. Doing the Gaussian integral gives

$$E(\alpha) \leq -\sqrt{\frac{\alpha}{\pi}} |V_0| \sqrt{\frac{\pi}{\alpha + 1/\Delta^2}} + \frac{\alpha \hbar^2}{4m} = -|V_0| \sqrt{\frac{\alpha}{\alpha + 1/\Delta^2}} + \frac{\alpha \hbar^2}{4m}$$

For $\alpha \rightarrow 0$ the right hand side becomes negative since the positive kinetic energy approaches zero faster than the negative potential energy.

Since from part (a) $E_0 \leq E(\alpha)$ irrespective of the value for α and from part (b) $E(\alpha) < 0$ for sufficiently large α , we know that $E_0 < 0$. So the lowest energy eigenstate is bound ($E_0 < V(\infty)$), hence the potential has at least one bound state. (Q.E.D.)

Exercise 5.3.1 (page 166)

$$\hat{H} = \frac{\hat{p}^2}{2m} + \hat{V}(\hat{x}) \quad \text{where} \quad V(x) = V_r - iV_i \quad (V_r, V_i \text{ real constants}).$$
$$= \frac{\hat{p}^2}{2m} + (V_r - iV_i)\hat{I}$$

$$\hat{H}^\dagger = \left(\frac{\hat{p}^2}{2m}\right)^\dagger + \hat{V}^\dagger = \frac{\hat{p}^2}{2m} + V_r\hat{I} + iV_i\hat{I} \neq \hat{H}$$

$\hat{H}$ is not Hermitian.

In this case

$$P(t) = \langle \psi(t) | \psi(t) \rangle = \langle \psi(0) | \hat{U}^\dagger(t) \hat{U}(t) | \psi(0) \rangle$$

$$= \langle \psi(0) | (e^{-i\hat{H}t/\hbar})^\dagger e^{-i\hat{H}t/\hbar} | \psi(0) \rangle$$

$$= \langle \psi(0) | e^{i\hat{H}^\dagger t/\hbar} e^{-i\hat{H}t/\hbar} | \psi(0) \rangle$$

$$= \langle \psi(0) | e^{i(iV_i)\hat{I}t/\hbar} e^{-i(-iV_i)\hat{I}t/\hbar} | \psi(0) \rangle$$

(since $\hat{p}^2$ commutes with $\hat{I}$ and with itself)

$$= e^{-2V_i t/\hbar} \langle \psi(0) | \psi(0) \rangle = e^{-\frac{2}{\hbar} V_i t} P(0)$$

$\Rightarrow P(t)$ decreases exponentially with time.

Exercise 5.3.4 (page 167)

Consider $\psi(x) = A e^{ipx/\hbar} + B e^{-ipx/\hbar}$.

Compute

$$\begin{aligned} j &= \frac{\hbar}{2mi} \left(\psi^* \frac{d\psi}{dx} - \psi \frac{d\psi^*}{dx} \right) \\ &= \frac{\hbar}{2mi} \left[(A^* e^{-ipx/\hbar} + B^* e^{ipx/\hbar}) \frac{ip}{\hbar} (A e^{ipx/\hbar} - B e^{-ipx/\hbar}) \right. \\ &\quad \left. - (A e^{ipx/\hbar} + B e^{-ipx/\hbar}) \frac{ip}{\hbar} (-A^* e^{-ipx/\hbar} + B^* e^{ipx/\hbar}) \right] \\ &= \frac{p}{2m} \left[(|A|^2 - \underbrace{A^* B}_{\cancel{A^* B}} e^{-2ipx/\hbar} + \underbrace{B^* A}_{\cancel{B^* A}} e^{2ipx/\hbar} - |B|^2) \right. \\ &\quad \left. - (-|A|^2 + \underbrace{AB^*}_{\cancel{AB^*}} e^{2ipx/\hbar} - \underbrace{BA^*}_{\cancel{BA^*}} e^{-2ipx/\hbar} + |B|^2) \right] \\ &= \frac{p}{m} (|A|^2 - |B|^2) \quad \checkmark \end{aligned}$$

Problem 5

(a) Angular momentum: $\hat{\vec{L}} = \hat{\vec{X}} \times \hat{\vec{P}} = (\hat{Y}\hat{P}_z - \hat{Z}\hat{P}_y, \hat{Z}\hat{P}_x - \hat{X}\hat{P}_z, \hat{X}\hat{P}_y - \hat{Y}\hat{P}_x)$

Since $\hat{X}, \hat{Y}, \hat{Z}, \hat{P}_x, \hat{P}_y, \hat{P}_z$ are all Hermitian and $\hat{X}$ commutes with $\hat{P}_y, \hat{P}_z$, $\hat{Y}$ commutes with $\hat{P}_x, \hat{P}_z$, and $\hat{Z}$ commutes with $\hat{P}_x, \hat{P}_y$, all 3 components of $\hat{\vec{L}}$ are Hermitian.

(b) $\frac{d}{dt} \langle \hat{\vec{L}} \rangle = -\frac{i}{\hbar} \langle [\hat{\vec{L}}, \hat{H}] \rangle$ Ehrenfest theorem

So we must compute $[\hat{\vec{L}}, \hat{H}]$.

$$\begin{aligned} \text{Start with } [\hat{L}_x, \hat{\vec{P}} \cdot \hat{\vec{P}}] &= [\hat{Y}\hat{P}_z - \hat{Z}\hat{P}_y, \hat{P}_x^2 + \hat{P}_y^2 + \hat{P}_z^2] \\ &= \hat{Y}[\hat{P}_z, \hat{P}_x^2 + \hat{P}_y^2 + \hat{P}_z^2] + [\hat{Y}, \hat{P}_x^2 + \hat{P}_y^2 + \hat{P}_z^2]\hat{P}_z \\ &\quad - \hat{Z}[\hat{P}_y, \hat{P}_x^2 + \hat{P}_y^2 + \hat{P}_z^2] - [\hat{Z}, \hat{P}_x^2 + \hat{P}_y^2 + \hat{P}_z^2]\hat{P}_y \\ &= \underbrace{([\hat{Y}, \hat{P}_y]\hat{P}_z + \hat{P}_y[\hat{Y}, \hat{P}_y])}_{i\hbar\hat{I}}\hat{P}_z - \underbrace{([\hat{Z}, \hat{P}_z]\hat{P}_x + \hat{P}_z[\hat{Z}, \hat{P}_z])}_{i\hbar\hat{I}}\hat{P}_x \\ &\quad - \underbrace{([\hat{Z}, \hat{P}_z]\hat{P}_y + \hat{P}_z[\hat{Z}, \hat{P}_y])}_{i\hbar\hat{I}}\hat{P}_y - \underbrace{([\hat{Y}, \hat{P}_y]\hat{P}_x + \hat{P}_y[\hat{Y}, \hat{P}_x])}_{i\hbar\hat{I}}\hat{P}_x \\ &= 2i\hbar(\hat{P}_y\hat{P}_z - \hat{P}_z\hat{P}_y) = 0 \end{aligned}$$

By inspection it is obvious that we will similarly find

$$[\hat{L}_y, \hat{\vec{P}} \cdot \hat{\vec{P}}] = [\hat{L}_z, \hat{\vec{P}} \cdot \hat{\vec{P}}] = 0$$

i.e. the angular momentum operator commutes with the kinetic energy operator.

Next we compute

$$\begin{aligned}
 [\vec{L}_x, \vec{X} \cdot \vec{X}] &= [\hat{Y}\hat{P}_z - \hat{Z}\hat{P}_y, \hat{X}^2 + \hat{Y}^2 + \hat{Z}^2] \\
 &= \hat{Y}[\hat{P}_z, \hat{X}^2 + \hat{Y}^2 + \hat{Z}^2] + [\hat{Y}, \hat{X}^2 + \hat{Y}^2 + \hat{Z}^2]\hat{P}_z \\
 &\quad - \hat{Z}[\hat{P}_y, \hat{X}^2 + \hat{Y}^2 + \hat{Z}^2] - [\hat{Z}, \hat{X}^2 + \hat{Y}^2 + \hat{Z}^2]\hat{P}_y \\
 &= -\hat{Y}[\hat{Z}^2, \hat{P}_z] + \hat{Z}[\hat{Y}^2, \hat{P}_y] \\
 &= -\hat{Y}\left(\underbrace{\hat{Z}[\hat{Z}, \hat{P}_z]}_{i\hbar\hat{I}} + \underbrace{[\hat{Z}, \hat{P}_z]\hat{Z}}_{i\hbar\hat{I}}\right) + \hat{Z}\left(\underbrace{\hat{Y}[\hat{Y}, \hat{P}_y]}_{i\hbar\hat{I}} + \underbrace{[\hat{Y}, \hat{P}_y]\hat{Y}}_{i\hbar\hat{I}}\right) \\
 &= 2i\hbar(-\hat{Y}\hat{Z} + \hat{Z}\hat{Y}) = 0
 \end{aligned}$$

Similarly $[\vec{L}_y, \vec{X}^2] = [\vec{L}_z, \vec{X}^2] = 0$, i.e. all components of $\vec{L}$ also commute with the potential energy operator.

In summary, $[\vec{L}, \hat{H}] = 0$, hence

$$\boxed{\frac{d}{dt} \langle \vec{L} \rangle = 0}$$

i.e. the expectation value of the angular momentum operator in any state is time independent

→ angular momentum conservation.